Rocket Science and Technology 4363 Motor Ave., Culver City, CA 90232
Phone: (310) 839-8956 Fax: (310) 839-8855

Kite Catenary

26 November 2006

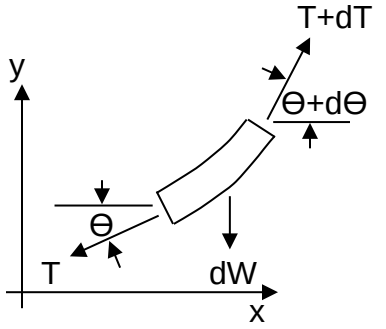
by C.P. Hoult

Introduction

While in kite mode the usual aircraft equilibrium equations must be modified to reflect the forces from the attached anchor line. Since the wind speed is low for the most interesting condition, minimum speed, we neglect aerodynamic drag on the anchor line. This implies the anchor line can be modeled as a classical catenary.

Catenary in Equilibrium

Consider an element of perfectly limp line in equilibrium under the influence of only gravity and axial tension as sketched below:



The conditions for force equilibrium of the x and y components are

$$\mathbf{(T + dT) \cos (\theta + d\theta) - T \cos \theta = 0}, \text{ and} \quad (1)$$

$$\mathbf{(T + dT) \sin (\theta + d\theta) - T \sin \theta = dW}, \text{ where} \quad (2)$$

T = Tension force in the line,
W = Weight of the line, and
θ = Slope of the line.

Expanding the trigonometric functions, simplifying and noting that

$$\mathbf{dW = \rho g\, ds}, \text{ where}$$

ρ = Mass of the line per unit length,
g = Acceleration of gravity, and
s = Distance along the line,

we obtain

$$\mathbf{d\,(\,T\cos\theta\,)/dx\;=0}\text{, and} \tag{3}$$

$$\mathbf{d\,(\,T\sin\theta\,)/dx\;=\;\rho\,g\,ds/dx.} \tag{4}$$

Next, recall that

$$\mathbf{\cos\theta\;=\;1\,/\,\sqrt{1+\;y'^2}}\text{, and}\quad \mathbf{\sin\theta\;=\;y'\,/\,\sqrt{1+\;y'^2}\,.}$$

As a consequence of eq. (3),

$$\mathbf{T\;=\;C\,\sqrt{1+\;y'^2},}$$

where C is a constant of integration. Substituting this into eq.(4), and simplifying,

$$\mathbf{d\,(C\,y')\,/\,dx\;=\;\rho\,g\,\sqrt{1+y'^2}}\,. \tag{5}$$

Noting that the origin of the **x,y** frame is at the catenary minimum (where $\mathbf{y'\;=\;0)}$, the first integral for eq. (5) is

$$\mathbf{y'\;=\;\sinh\,(\,\rho\,g\,x\,/\,C\,)}, \tag{6}$$

and, the line tension is

$$\mathbf{T\;=\;C\cosh\;\,(\,\rho\,g\,x\,/\,C\,)}. \tag{7}$$

The second integral for eq. (5) is

$$\mathbf{y\;=\;(\,C\,/\,\rho\,g)\,[\,\cosh\;\,(\,\rho\,g\,x\,/\,C\,)-1\,]}. \tag{8}$$

Finally, eq. (4) can be construed to state that the vertical component of line tension at any x value must equal the weight of the line below it. Then,

$$\mathbf{T\sin\theta\;=\;\rho\,g\,\Lambda\;=\;C\sinh\;\,(\,\rho\,g\,x\,/\,C\,)}\text{, or}$$

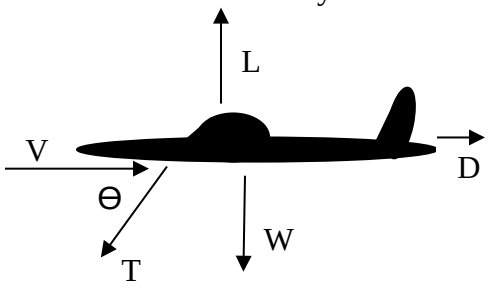
$$\mathbf{\rho\,g\,\Lambda\,/\,C\;=\;\sinh\;\,(\,\rho\,g\,x\,/\,C\,)}. \tag{9}$$

where $\mathbf{\Lambda}$ = Length of line from the origin to x.

Equation (9) is a simple transcendental result that can readily be solved. Since many of the practical aerospace applications of catenaries use specified line lengths, use eq's. (9) and (8) to establish the family of catenary shapes.

<u>Aircraft Kite in Equilibrium</u>

Next, consider an aircraft kite in static equilibrium. The four forces acting on an aircraft kite are aerodynamic lift and drag, weight and tension from the catenary line. The force equilibrium eq's are:



$\mathbf{D = T \cos \Theta = T / \sqrt{1 + y'^2}}$, and

$\mathbf{L = W + T \sin \Theta = W + T\, y' / \sqrt{1 + y'^2}}$.

Substituting from eq's.(6) and (7) leads to

$\mathbf{D = C}$, and

$$\mathbf{L = W + D \sinh (\rho g x / D)}. \qquad (10)$$

Now, consider an algorithmic implementation. First, define two points along an imaginary catenary. Point "0" is at the top of the catenary where it attaches to the aircraft kite. Point "1" is at the ground anchor point. From eq. (10), the value of $\mathbf{x_0}$ is found to be

$$\mathbf{x_0 = (D / \rho g) \sinh^{-1} ((L - W) / D)}.$$

Next, find $\mathbf{x_1}$ for a specified line length $\Delta\Lambda$ from anchor to aircraft:

$$\Delta\Lambda = \Lambda_0 - \Lambda_1, \text{ or}$$

$$\Delta\Lambda = (D / \rho g) [\sinh (\rho g x_0 / D) - \sinh (\rho g x_1 / D)], \text{ or}$$

$$x_1 = (D / \rho g) \sinh^{-1} ((L - W - \rho g \Delta\Lambda) / D).$$

The aircraft equilibrium altitude is found from eq. (8)

$$y_0 - y_1 = (D / \rho g) [\cosh (\rho g x_0 / D) - \cosh (\rho g x_1 / D)].$$

Last, the line tension is greatest at the aircraft attachment point, and is given by

$$T_{max} = D \cosh (\rho g x_0 / D). \qquad (11)$$

Tethered Pibal

In sounding rocket parlance, the word "pibal" is short hand for pilot balloon. Pibals (typically a 1 meter helium-filled neoprene rubber sphere) are launched at intervals prior to a sounding rocket launch. Because they drift laterally with the wind, a sequence of position estimates taken on a pibal as it ascends can be used to estimate the vector wind profile as a function of altitude. Knowing the wind profile the sounding rocket trajectory can be predicted and controlled.

For financially constrained amateur sounding rocket operations, the expense of operating dual tracking theodolites and multiple balloon launches per sounding rocket launch can prove excessive. As a counter, I suggest the use of a few tethered balloons observed with

a single theodolite located near the tether anchor point. The observed data consist of wind azimuth and balloon zenith angle. These two parameters can be used to estimate one point on the vector wind profile as a function of altitude. Other points can be found from additional tethered balloons. The thing is, the balloons can remain in place all day, and can, up to a point, be reused.

The tethered pibal works just like the aircraft except the aerodynamic lift is replaced by the static buoyant lift. However, there is one important difference: Most aircraft kites are flown with a tether of fixed length while most pibals are planned to be flown at fixed, specified altitude, **H**. Then

$$H = (D / \rho g) [\cosh (\rho g x_0 / D) - \cosh (\rho g x_1 / D)].$$

Also,

$$x_0 = (D / \rho g) \sinh^{-1} ((L - W) / D), \text{ and}$$

$$x_1 = (D / \rho g) \sinh^{-1} ((L - W - \rho g \Delta\Lambda) / D).$$

Solve the first eq. for x_1:

$\rho g x_1 / D = \cosh^{-1} [\cosh (\rho g x_0 / D) - \rho g H / D]$, and substitute into the third eq.:

$$\cosh^{-1} [\cosh (\rho g x_0 / D) - \rho g H / D] = \sinh^{-1} ((L - W - \rho g \Delta\Lambda) / D).$$

Next, substitute for x_0 from the second eq.:

$\cosh^{-1} [\cosh (\sinh^{-1} ((L - W) / D)) - \rho g H / D] = \sinh^{-1} ((L - W - \rho g \Delta\Lambda) / D)$, or

$$\cosh^{-1} [\sqrt{1 + ((L - W) / D))^2} - \rho g H / D] = \sinh^{-1} ((L - W - \rho g \Delta\Lambda) / D).$$

$$\sqrt{1 + ((L - W) / D))^2} - \rho g H / D = \sqrt{((L - W - \rho g \Delta\Lambda) / D)^2 + 1}, \text{ or}$$

$$(\rho g H / D)^2 - 2 \rho g H \sqrt{1 + ((L - W) / D))^2} / D = 2 \sqrt{1 + ((L - W) / D))^2} x$$

$$(\rho g \Delta\Lambda) / D + (\rho g \Delta\Lambda) / D)^2$$

(12)

Equation (12) is quadratic in the necessary tether length, $\Delta\Lambda$, and can be solved in the usual way.

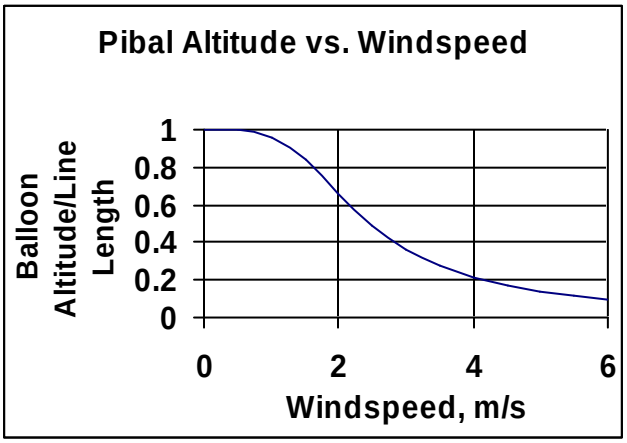
An Approximation

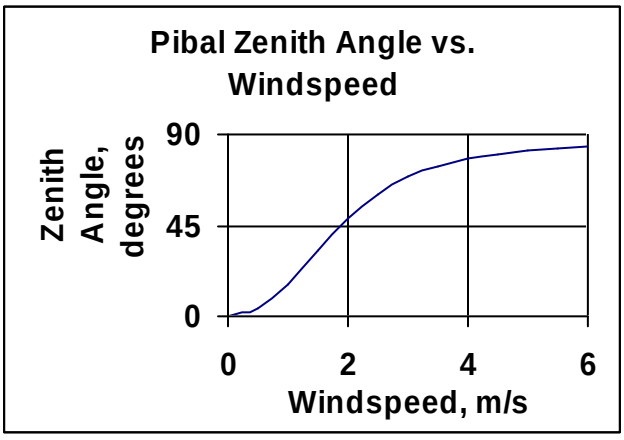
A good first approximation is that for a weightless line. In this approximation, the tether is a straight from ground to pibal. Then, the static equilibrium equations are

D = T sin θ, or T = D / sin θ, and **L – W = T cos θ**, or **tan θ = D / L – W**, and the tether length is **ΔΛ = H / cos θ**.

The zenith angle ζ can be found from

$$\zeta = \tan^{-1} [(x_0 - x_1) / (y_0 - y_1)]. \quad (13)$$

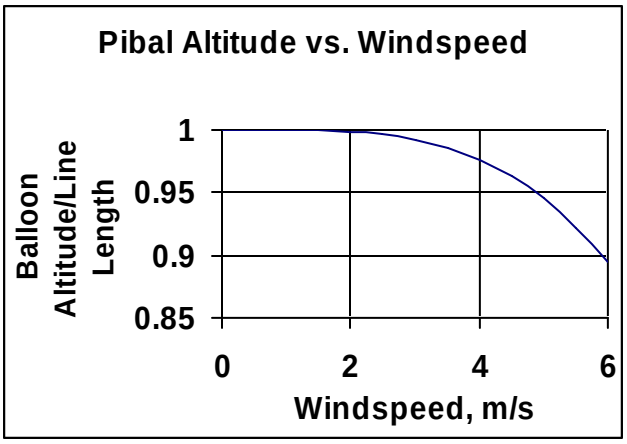




Consider a typical example of a 1 meter neoprene rubber weather balloon tethered with 100 m of 0.007" Spectra 2000 line. The result is sketched on the left. First, the Spectra line is so strong that a weightless line approximation works well. The breaking stress of the tether is not reached until the windspeed is about 5.7 m/s.

As a corollary, the zenith angle vs. windspeed is essentially universal…it is independent of line length. And, lastly, the balloon altitude curve is also universal in the same sense.

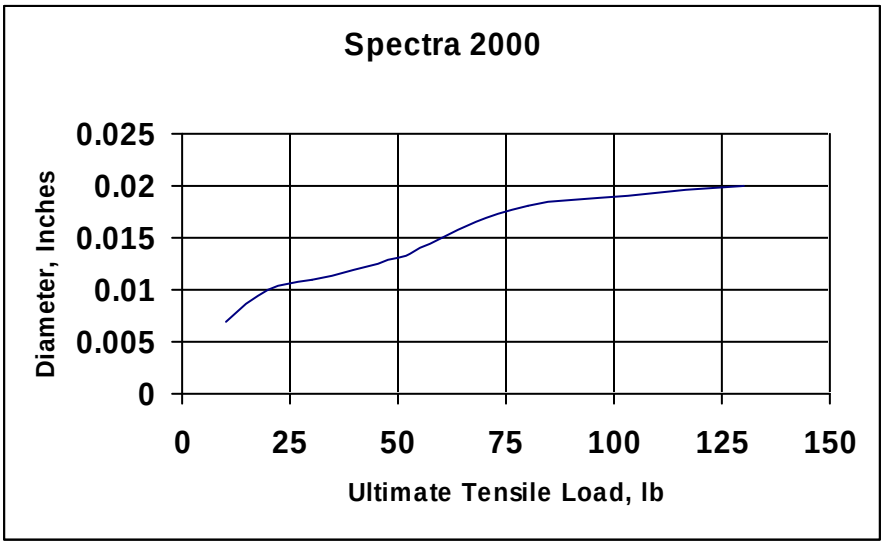
The catch is that a greater lifting force can make a big difference. The bottom sketch shows the result for an artificial balloon with 10 kg of net lift, about 20 times that available from a real weather balloon. But, the line tension at 5.7 m/s has only increased from 10 lbs to ~23 lbs.



Spectra® 2000

Spectra® 2000 is the latest in high strength synthetic fibers. It is a carefully formulated polyethylene literally ten times stronger than steel on a pound for pound basis. Its small diameter makes it hard to detect visually, and its low dielectric constant makes it nearly invisible to radar. Marketed by Honeywell International, it is used for kite line, fishing line (PRT Braid), bullet-proof vests, etc. Its density of only 0.97 gr/cc means it can float on water. If we need to cut it to depart kite mode, it can be electrically melted at 150º C.

The figure below shows line diameter as a function of ultimate tensile strength.



Spectra® 2000 PRT Braid